



Leica DFC495

**High Performance Digital FireWire Color Camera System
for High-resolution Photomicrography**

Living up to Life

Leica
MICROSYSTEMS

Ultra High-resolution Phot

Feature Highlights:

- 8 Megapixel CCD for excellent, high-resolution images
- Quick transfer to PC with standard FireWire 1394b interface
- SXGA 1208×960 progressive scan preview with up to 16 frames per second
- 36 bit RGB color depth
- Peltier cooling for high dynamic range and minimum noise in low light situations
- Exposure times from 2 msec to 600 sec
- Fastest scanning of a freely defined area at full resolution
- Simple, fast connection to all microscopes via the C-mount interface
- Excellent live image mode for quick focusing and positioning
- Power supply and quick, reliable data transfer with only one cable
- Intuitive user interface with convenient image capture and processing functions for PCs
- Outstanding image quality
- Two-color LED displays operation status

New applications in life science and industry require innovative approaches to imaging. Quickly producing high-quality images for documentation, evaluation, and analysis is the key to imaging success. The Leica DFC495 digital camera system provides the highest color fidelity, resolution, and detail. Real time speeds can be achieved using a variety of innovative readout modes. The Leica DFC495 is ideal for the most intricate documentation.

High-resolution Detail

The cutting edge Leica DFC495 integrates an 8-megapixel CCD, offering superior quality, high-resolution images that were previously only possible with multiple acquisition cameras. High-resolution CCD's are especially beneficial for low magnification microscope imaging, as the amount of information provided by the optical system is much larger than with high magnification. The DFC495 produces sharp images with unsurpassed color accuracy that represents the best in the industry.

Innovative Readout

Innovative data readout modes allow the user to freely select image transfer, speed, and scan method. Full frame readout mode, for example, utilizes full camera resolution (3264×2448 pixels) with 12-bit signal processing uncompressed TIFF files and producing 38 MB files. By saving such huge image data in 3×8 Bit only and in JPEG format, you can of course reduce the image size considerably.

Uniquely Engineered

The Leica DFC495 incorporates a three-tiered cooling system designed to eliminate thermal noise buildup in the camera. An integrated Peltier cooling system draws heat away from the CCD, eliminating excited electrons in the camera head. Leica's unique metallic camera housing with heat-dissipating fins pulls heat away from the camera. The DFC495, like all Leica cameras, uses the photo coupler as a heat dissipation conduit.



ography Provides Precise detail

System Integration

Creating crisp, sharp images was never easier with the new Leica Application Suite (LAS) software. LAS features automatic microscope setup and calibration, annotation, and measurement functions. If LAS is used with an automated microscope, the user can store and recall camera and microscope parameters to exactly reproduce previously made pictures. The unique workflow approach of LAS makes processing and organizing high-resolution images a snap.

High-performance Leica LAS software

The Leica Application Suite software (LAS) included in the scope of supply offers numerous functions for recording and retouching images. Beginners as well as experienced users can thus use the full potential of the digital technology. The captured images can be edited, analysed, archived and reproduced as often as you wish without any loss in quality.



Leica DM4000M with Leica DFC495 Digital Microscope Camera and PC System with Leica LAS Software

Technical Data: Leica DFC495

Digital Camera		Leica DFC495	
Camera type	Digital camera for microscopy with control software		
Sensor	Interline transfer frame readout CCD – ICX456		
Sensor Grade/Size	Grade Zero / 8.81 mm × 6.61 mm (Type 2/3")		
Color filter	RGB Bayer mosaic		
Protective color filter	Hoya CM500S (IR cut-off 650 nm), removable		
Shutter control	Electronic global shutter/interlaced readout, 3 images		
Number of pixels / Pixel size	8 Megapixel, 3264 × 2448 / 2.7 μm × 2.7 μm		
Color depth	36 Bit		
A/D converter	14 Bit		
Dynamic range	> 58 dB / > 800:1		
Exposure time	2 msec – 600 sec		
Readout noise	σ < 6 LSB (12 bit) typical		
Gain control/Offset control	10× / 0.. 255 LSB (12 bit)		
Shading correction	Yes, stored for all formats		
Cooling	Δ –20° to ambient		
Region of interest	Freely adjustable in 2 pixels steps from 2 × 2 up to full resolution		
Image Formats		Pixel	Images per Second, Fast / HQ
Interlaced large		3264 × 2448	4/2
Interlaced medium		2176 × 1632	6/3
Progressive large (SXGA)		1280 × 960	16/ 8
Progressive medium		1088 × 816	24/ 12
Progressive small		544 × 408	48/ 24
Modes		Formats in Fast (40MHz) or High Quality (20MHz) modes	
Computer	Hardware		Software
Min. computer configuration	Pentium 4, 2.5 GHz, 1 GB RAM 24 bit graphics, 1024 × 768, 6-pin or 9-pin FireWire OHCI or free PCI slot		Leica DFC Twain Leica LAS Software Windows Vista 32/64 bit Windows XP prof. 32/64 bit, SP3
Interfaces			
Recommended video adapter		C-Mount 0.63× or 0.8×	
Data		Single cable FireWire – IEEE1394b 9-pin, screw-lock	
Digital Input connector		Opto-decoupled trigger	
Digital Output connector		Flash synch or readout active	
Physical and Environmental			
Power consumption		~6 W	
Power supply		via FireWire cable	
Housing		Aluminium die cast	
Size		132 × 74 × 69 mm³	
Weight		495 g	
Operating temperature range		+5 to +35 °C	
Relative humidity		10 % .. 80 % non condensing	
Order Numbers			
12 730 223		Leica DFC495 Camera kit comprising: Leica DFC495 Camera, Leica software, FireWire cable b–b	
Additional Items			
12 730 210		PCI-Express FireWire-b card for PCs without FireWire	
12 447 066		Laptop PCMCIA FireWire-a interface card	
12 730 186		FireWire cable, 3 m, b–b, 9/9-Pin	
12 730 187		FireWire cable, 3 m, a–b, 6/9 Pin	
12 730 188		FireWire Power kit comprising: 110/220V power supply, 4-pin FireWire-a or 6-pin FireWire-a adapter	

www.leica-microsystems.com

